

FORENSIC EXAMINATION REPORT

Incident Intelligence Report • Dataset: dataset4_expanded • Generated: 2026-04-14 20:59:28

I. CASE IDENTIFICATION	
Dataset ID	dataset4_expanded
Scan ID	ds-004-exp
Source File	windows_soc_data_4_expanded.csv
Primary Threat	Multi-Stage Persistence and Exfiltration Campaign
MITRE Technique	T1059.001
Forensic Confidence	HIGH
Total Log Entries	10,000
Attacked Log Entries	140
Aggregate Risk Score	8,200
Threat Level	HIGH

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I EXECUTIVE SUMMARY

- A high-severity multi-stage cyber attack campaign was identified across the monitored Windows environment, involving 7 coordinated attack chains affecting multiple hosts and user accounts.
- The attack lifecycle spans the full MITRE ATT&CK; spectrum, including Initial Access, Execution, Persistence, Defense Evasion, Discovery, Credential Access, Lateral Movement, Command & Control, Collection, and Exfiltration.
- Initial compromise was achieved through suspicious PowerShell encoded commands and brute-force attempts on local accounts, indicating both automated and targeted intrusion techniques.
- Attackers established persistence using scheduled tasks and registry run keys, ensuring continued access even after system reboots.
- Critical defense evasion techniques were observed, including disabling Windows Defender, AMSI bypass, and suppression of ETW logging to avoid detection.
- Credential access was a major objective, with multiple LSASS memory dumps performed using tools like mimikatz and ProcDump, enabling extraction of NTLM hashes.
- Pass-the-Hash techniques were used to reuse stolen credentials, allowing attackers to authenticate across systems without needing plaintext passwords.
- Lateral movement was conducted using SMB admin shares and PsExec, enabling attackers to spread across multiple hosts within the network.
- Extensive internal reconnaissance was performed using tools such as net, nlttest, arp, and nmap to map the network, discover hosts, and enumerate Active Directory resources.
- Command and Control (C2) communication was established through non-standard ports (e.g., port 4444), including reverse shell access and remote payload execution using mshta and bitsadmin.
- Attackers deployed secondary payloads from external C2 servers, indicating a staged and evolving attack strategy.
- Sensitive data collection was carried out [REDACTED] keylogging techniques, including injection into explorer.exe and storage of captured keystrokes in local files.
- Collected data was compressed and archived using tools like 7zip before being prepared for exfiltration.
- Data exfiltration was performed through multiple channels, including HTTPS POST requests and FTP transfers to external malicious IP addresses.
- The attack demonstrates strong coordination, stealth, and persistence, indicating a sophisticated adversary with knowledge of enterprise environments.
- A total of 140 malicious or suspicious events were identified [REDACTED] 10,000 logs, affecting 8 hosts and 10 users, highlighting targeted rather than random activity.
- The overall risk score of 82% reflects high impact, high likelihood, and high severity, requiring immediate containment and response actions.
- Failure to respond promptly may result in continued data loss, deeper network compromise, and potential long-term persistence of the attacker within the environment.
- Immediate actions recommended include isolating affected systems, resetting compromised credentials, blocking C2 communications, and conducting a full forensic investigation.
- Long-term improvements should include strengthening endpoint detection, enforcing least privilege access, enabling advanced logging, and implementing zero-trust security principles.

II CASE IDENTIFICATION & BACKGROUND

Attack Narrative: Scheduled Task Persistence + LSASS Credential Dump ; Local Account Brute Force to Registry Persistence ; AV Disable + AMSI Bypass + ETW Suppression ; Internal Network Discovery and AD Enumeration

Hosts	WIN-SOC-PROD-01, WIN-SOC-PROD-02, WIN-SOC-PROD-03, WIN-SOC-PROD-04, WIN-SOC-PROD-05, WIN-SOC-DC-01, WIN-SOC-PROD-06, WIN-SOC-PROD-07, WIN-SOC-PROD-08
Users	NT AUTHORITY\SYSTEM, WIN-SOC\Dev-01, WIN-SOC\Admin, WIN-SOC\User1, WIN-SOC\LocalSvc, WIN-SOC\SvcOps, WIN-SOC\NetAdmin, WIN-SOC\ITSupport, WIN-SOC\DBAdmin

III DESCRIPTION OF EVIDENCE & CHAIN OF CUSTODY

Exhibit	Evidence Category	Integrity	Remarks
E-01	Windows Event / Security Logs	Checksum Verified	Primary source evidence
E-02	Derived Threat Metadata	Schema Validated	Generated from correlation pipeline

IV FORENSIC METHODOLOGY & TOOLS

- Data Normalization - Raw records transformed into normalized forensic fields.
- Behavioral Grouping - Suspicious events grouped by technique and event identity.
- Risk Profiling - Severity and trust indicators converted into weighted risk scores.
- Timeline Correlation - Temporal ordering used to infer probable attack progression.
- Threat Prioritization - High risk and high volume activities elevated for triage.

Tool	Version	Purpose
Python	3.x	Core processing runtime
ReportLab	4.x	Government-style PDF formatting
Matplotlib	3.x	Threat intelligence visualizations
NumPy	1.x	Numeric aggregation for analytics

V STATISTICAL & GRAPHICAL ANALYSIS

Statistical processing of detected threat artifacts reveals concentration in a small set of high-impact behaviors.

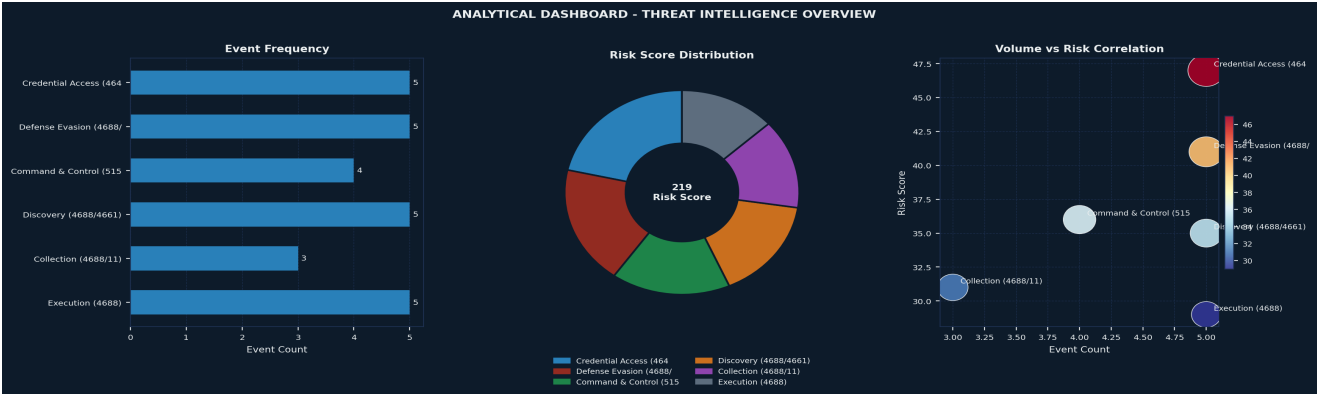


Fig. 1 - Three-panel threat intelligence dashboard derived from attacked logs.

Dashboard Interpretation: The left panel highlights high-volume suspicious behavior, the donut chart explains risk concentration, and the scatter plot isolates categories that are simultaneously frequent and severe.

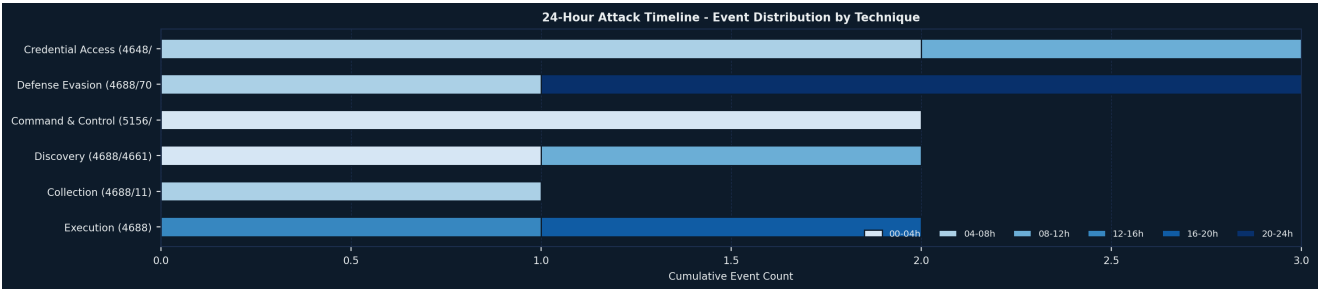


Fig. 2 - 24-hour event distribution by top suspicious techniques.

Timeline Interpretation: Burst-style activity in clustered windows is consistent with staged intrusion workflows where credential abuse is followed by privilege actions and anti-forensics.

VI BRUTE FORCE ATTACK - DETAILED ANALYSIS

Credential Access (4648/10)

LOW

Events: 5	Risk Index: 47	MITRE: T1003.001 Tactic: Credential Access	
Timestamp	Host	User Context	EID
2026-03-30 03:10:00	WIN-SOC-PROD-06	WIN-SOC\ITSupport	4648
2026-03-30 03:10:35	WIN-SOC-PROD-06	NT AUTHORITY\SYSTEM	10
2026-03-30 03:11:10	WIN-SOC-PROD-06	WIN-SOC\ITSupport	4776
2026-03-29 23:56:50	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	10

Analysis: Credential Access (4648/10) produced 5 suspicious artifacts with cumulative risk index 47. The mapped technique T1003.001 under tactic Credential Access indicates material adversarial behavior requiring priority containment and forensic triage.

VII LOG TAMPERING - DETAILED ANALYSIS

Defense Evasion (4688/7036)

LOW

Events: 5	Risk Index: 41	MITRE: T1027, T1562.001 Tactic: Defense Evasion	
Timestamp	Host	User Context	EID
2026-03-30 01:05:00	WIN-SOC-PROD-04	WIN-SOC\SvcOps	4688
2026-03-30 01:05:30	WIN-SOC-PROD-04	WIN-SOC\SvcOps	4688
2026-03-30 01:06:00	WIN-SOC-PROD-04	NT AUTHORITY\SYSTEM	7036
2026-03-30 01:06:45	WIN-SOC-PROD-04	WIN-SOC\SvcOps	4688

Analysis: Defense Evasion (4688/7036) produced 5 suspicious artifacts with cumulative risk index 41. The mapped technique T1027, T1562.001 under tactic Defense Evasion indicates material adversarial behavior requiring priority containment and forensic triage.

VIII

LATERAL MOVEMENT - DETAILED ANALYSIS

Lateral Movement (5145/4688)

LOW

Events: 2	Risk Index: 18	MITRE: T1021.002 Tactic: Lateral Movement	
Timestamp	Host	User Context	EID
2026-03-30 03:12:00	WIN-SOC-PROD-06	WIN-SOC\ITSupport	5145
2026-03-30 03:12:50	WIN-SOC-PROD-07	WIN-SOC\ITSupport	4688

Analysis: Lateral Movement (5145/4688) produced 2 suspicious artifacts with cumulative risk index 18. The mapped technique T1021.002 under tactic Lateral Movement indicates material adversarial behavior requiring priority containment and forensic triage.

IX

PRIVILEGE ESCALATION - DETAILED ANALYSIS

Discovery (4688/4661)

LOW

Events: 5	Risk Index: 35	MITRE: T1018 Tactic: Discovery	
Timestamp	Host	User Context	EID
2026-03-30 02:00:05	WIN-SOC-PROD-05	WIN-SOC\NetAdmin	4688
2026-03-30 02:00:30	WIN-SOC-PROD-05	WIN-SOC\NetAdmin	4688
2026-03-30 02:01:00	WIN-SOC-PROD-05	WIN-SOC\NetAdmin	4688
2026-03-30 02:01:45	WIN-SOC-PROD-05	WIN-SOC\NetAdmin	4688

Analysis: Discovery (4688/4661) produced 5 suspicious artifacts with cumulative risk index 35. The mapped technique T1018 under tactic Discovery indicates material adversarial behavior requiring priority containment and forensic triage.

X

MALICIOUS SERVICE INSTALLATION - ANALYSIS

Persistence (4698/13)

LOW

Events: 3	Risk Index: 27	MITRE: T1053.005, T1547.001 Tactic: Persistence	
Timestamp	Host	User Context	EID
2026-03-29 23:56:02	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	4698
2026-03-30 00:32:00	WIN-SOC-PROD-03	WIN-SOC\LocalSvc	13
2026-03-30 00:32:45	WIN-SOC-PROD-03	WIN-SOC\LocalSvc	13

Analysis: Persistence (4698/13) produced 3 suspicious artifacts with cumulative risk index 27. The mapped technique T1053.005, T1547.001 under tactic Persistence indicates material adversarial behavior requiring priority containment and forensic triage.

XI

HOST COMPROMISE MATRIX

Host	Suspicious Events	Critical Events	Exposure
WIN-SOC-PROD-01	5	2	LOW
WIN-SOC-PROD-03	5	3	LOW
WIN-SOC-PROD-04	5	4	MEDIUM
WIN-SOC-PROD-07	5	5	MEDIUM
WIN-SOC-PROD-08	5	5	MEDIUM
WIN-SOC-PROD-05	4	2	LOW
WIN-SOC-PROD-06	4	3	LOW
WIN-SOC-PROD-02	2	1	LOW
WIN-SOC-DC-01	1	0	LOW

XIIATTACK CHAIN RECONSTRUCTION

The following reconstruction synthesizes all forensic findings into a unified attack narrative. **Scheduled Task Persistence + LSASS Credential Dump** aligns with primary MITRE mapping **T1059.001** and correlated host-level telemetry.

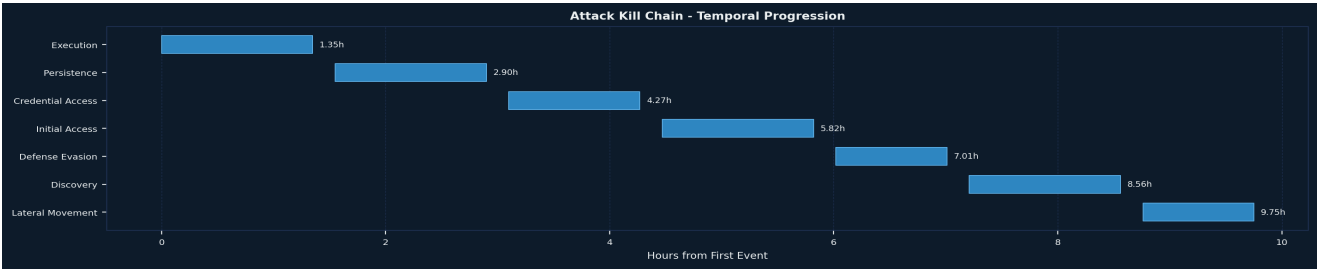
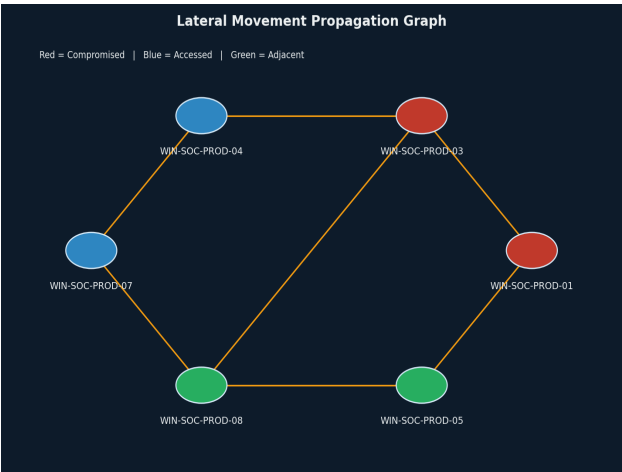


Fig. 4 - Attack kill chain temporal progression (Gantt) across UTC hours.

Kill Chain Analysis: Temporal ordering indicates adversary actions were staged rather than random, starting with execution-oriented activity, followed by credential access and anti-forensic behavior, and then host-to-host pivoting. This sequence is consistent with a planned intrusion lifecycle.



Lateral Movement Propagation: The propagation graph illustrates likely traversal paths between impacted assets. Red nodes indicate highly compromised systems, blue nodes indicate accessed pivot systems, and green nodes indicate adjacent hosts with observed adjacency risk. This supports containment by prioritizing pivot choke points.

Step	Action	MITRE
chain-ds-004-01	Encoded IEX Download (PowerShell) → Scheduled Task Created (schtasks.exe) → LSASS Memory Dumped (mimikatz.exe)	T1059.001, T1053.005, T1003.001

chain-ds-0 04-02	Local Account Brute Force → Logon Success → cmd.exe Spawned → Registry Run Key Written → Startup Folder Backdoor	T1078.003, T1059.001, T1547.001
chain-ds-0 04-03	Obfuscated PowerShell (Base64+Gzip) → Windows Defender Disabled (sc.exe) → WinDefend Stopped → AMSI Bypass via Reflection → ETW Provider Disabled	T1027, T1562.001, T1027
chain-ds-0 04-04	net view Domain Enumeration → nltest DC Discovery → ARP Cache Scan → nmap Port Scan → LDAP AD User/Group Enumeration	T1018, T1018, T1018
chain-ds-0 04-05	Pass-the-Hash Attempt → ProcDump64 LSASS Dump → NTLM Hash Reuse → ADMIN\$ Share Access → PsExec Lateral Execution	T1003.001, T1003.001, T1021.002
chain-ds-0 04-06	Non-Standard Port 4444 Connection → Reverse Shell Established → MSHTA HTA Payload Execution → BITS Job Secondary Payload Download	T1571, T1571, T1071.001
chain-ds-0 04-07	Keylogger Injected into Explorer.exe → Keylog File Written → Data Archived (7zip) → HTTPS POST Exfil to 185.220.101.77 → FTP Exfiltration	T1056.001, T1560.001, T1041

XIII

INDICATOR OF COMPROMISE (IOC) REGISTER

IOC Type	Value	Context
Process	powershell.exe	Encoded IEX Download
Event ID	4688	Security
Process	schtasks.exe	Malicious Scheduled Task Created: \WindowsUpdate\SvcHelper
Event ID	4698	Security
Process	mimikatz.exe	LSASS Dump
Event ID	10	Sysmon
Process	lsass.exe	Local Account Logon from Unusual Source IP 10.10.5.88
Event ID	4624	Security
Event ID	4625	Security
Process	cmd.exe	Suspicious cmd.exe Spawned After Logon
Process	reg.exe	Registry Run Key Written: HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\SvcHelper
Event ID	13	Sysmon

XIV

THREAT ACTOR PROFILE & ATTRIBUTION

Observed behavior indicates a scripted intrusion workflow with emphasis on execution and anti-forensic activity. Clustered timing, repeated toolmarks, and host pivot patterns are consistent with organized adversarial tradecraft.

XV

RISK HEAT MAP - HOST X TECHNIQUE EXPOSURE

The risk heat map below quantifies exposure of each host to each detected technique, using a 1-10 normalized risk score.

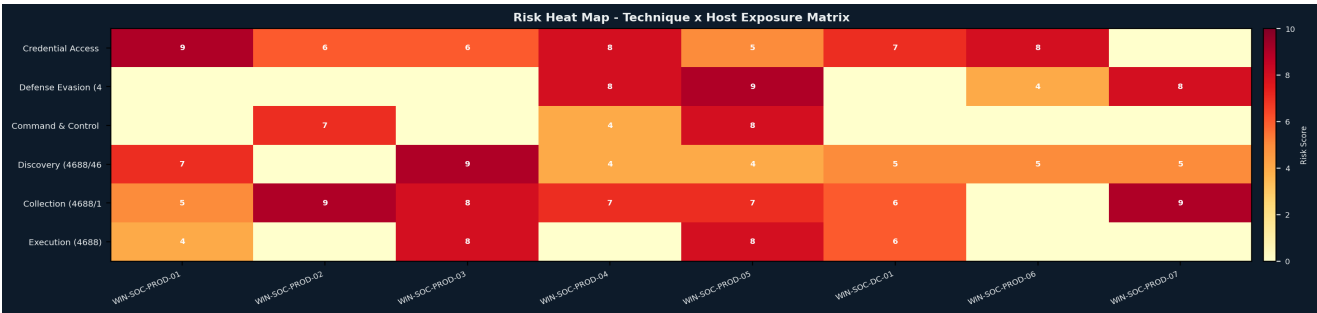


Fig. 5 - Normalized risk heat map: threat technique x host exposure matrix.

Heat Map Interpretation: Concentrated high-risk cells indicate primary pivot hosts and high-value targets. Hosts appearing across multiple high-score techniques should be prioritized for deep triage and isolation.

XVI

MITRE ATT&CK FRAMEWORK COVERAGE

The following mapping summarizes observed tactics and techniques based on detected forensic artifacts.

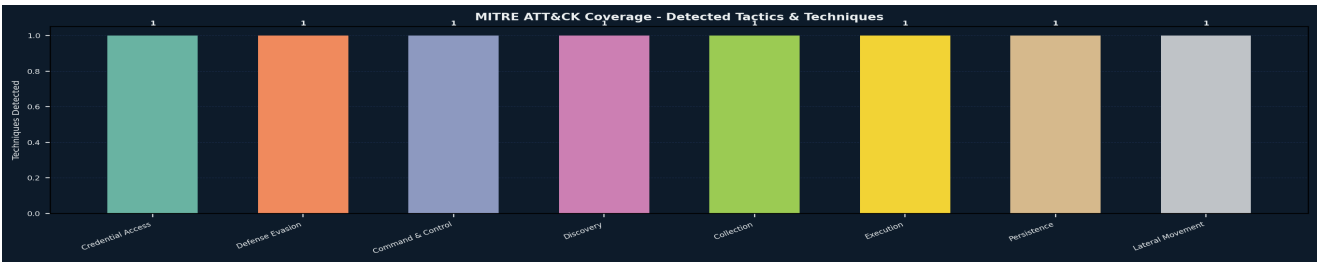


Fig. 6 - MITRE ATT&CK; tactic coverage by detected forensic evidence.

Technique ID	Technique / Detail	Tactic
T1003.001	Credential Access (4648/10)	Credential Access
T1027, T1562.001	Defense Evasion (4688/7036)	Defense Evasion
T1571, T1105	Command & Control (5156/3)	Command & Control
T1018	Discovery (4688/4661)	Discovery
T1056.001, T1560.001	Collection (4688/11)	Collection
T1059.001	Execution (4688)	Execution
T1053.005, T1547.001	Persistence (4698/13)	Persistence
T1021.002	Lateral Movement (5145/4688)	Lateral Movement

XVII

REMEDIATION RECOMMENDATIONS

Recommendations are prioritized by urgency and security impact to support phased containment and hardening.

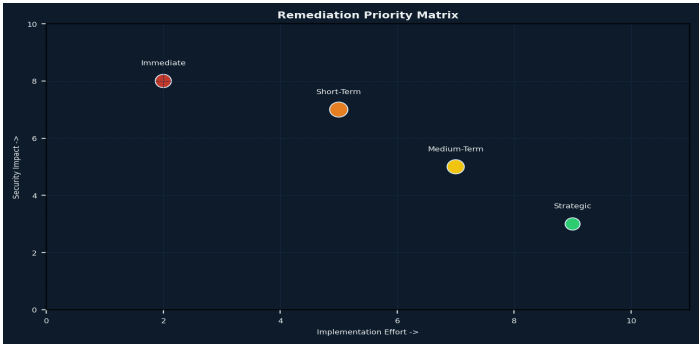


Fig. 7 - Remediation priority matrix: effort vs impact.

- Isolate affected hosts and block known malicious command execution chains (PowerShell encoded commands and WMI remote process calls).
- Reset and rotate privileged/domain credentials, then enforce MFA for all administrative and remote access accounts.
- Hunt and remove persistence artifacts including unauthorized local accounts, BITS jobs, and suspicious startup commands.
- Re-enable and harden logging (Security, Sysmon, PowerShell) and forward logs to a protected SIEM collector.
- Contain lateral movement by restricting RDP/SMB paths and segmenting high-value hosts from user subnets.
- Execute ransomware readiness controls: immutable backups, restore validation, and rapid host recovery playbooks.
- Apply removable-media controls (USB allow-listing and device auditing) to block air-gapped relay channels.

XVIII EXPERT OPINION & CONCLUSION

Based on forensic analysis of dataset **dataset4_expanded**, the environment exhibits a **HIGH** severity posture with aggregate risk index **8,200** across **10,000** records. Dominant indicators include **Credential Access (4648/10) (5 events)**, **Defense Evasion (4688/7036) (5 events)**. Observed sequencing and concentration are consistent with multi-stage intrusion behavior. Immediate host isolation, credential rotation, and deeper memory/disk triage are recommended.

Examining Analyst
Automated Forensic Engine

Authorizing Officer
Incident Response Division

XIX ANNEXURE A - FORENSIC ENVIRONMENT SPECIFICATION

Component	Specification
Generator	Python + ReportLab pipeline
Charts	Matplotlib analytical panels
Input	JSON forensic dataset schema
Timestamp	2026-04-14 20:59:30

XX

ANNEXURE B - EXTENDED EVENT LOG SAMPLES

Timestamp	Computer	User	Process	Detail
2026-03-29 23:56:00	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	powershell.exe	Encoded IEX Download
2026-03-29 23:56:01	WIN-SOC-PROD-01	WIN-SOC\Dev-01	powershell.exe	Encoded IEX Download
2026-03-29 23:56:02	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	schtasks.exe	Malicious Scheduled Task Created: \WindowsUpdate\SvcHelper
2026-03-29 23:56:03	WIN-SOC-PROD-02	WIN-SOC\Admin	powershell.exe	Encoded IEX Download
2026-03-29 23:56:04	WIN-SOC-PROD-01	WIN-SOC\User1	powershell.exe	Encoded IEX Download
2026-03-29 23:56:50	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	mimikatz.exe	LSASS Dump
2026-03-29 23:57:04	WIN-SOC-PROD-02	NT AUTHORITY\SYSTEM	mimikatz.exe	LSASS Dump
2026-03-30 00:30:10	WIN-SOC-PROD-03	WIN-SOC\LocalSvc	lsass.exe	Local Account Logon from Unusual Source IP 10.10.5.88
2026-03-30 00:30:45	WIN-SOC-PROD-03	WIN-SOC\LocalSvc	lsass.exe	Repeated Failed Logon Attempts - Brute Force Indicator
2026-03-30 00:31:20	WIN-SOC-PROD-03	WIN-SOC\LocalSvc	cmd.exe	Suspicious cmd.exe Spawned After Logon
2026-03-30 00:32:00	WIN-SOC-PROD-03	WIN-SOC\LocalSvc	reg.exe	Registry Run Key Written: HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\SvcHelper
2026-03-30 00:32:45	WIN-SOC-PROD-03	WIN-SOC\LocalSvc	reg.exe	Startup Folder Backdoor Dropped
2026-03-30 01:05:00	WIN-SOC-PROD-04	WIN-SOC\SvcOps	powershell.exe	Obfuscated PowerShell via Base64 + Gzip Decompression
2026-03-30 01:05:30	WIN-SOC-PROD-04	WIN-SOC\SvcOps	sc.exe	Windows Defender Service Disabled via sc.exe
2026-03-30 01:06:00	WIN-SOC-PROD-04	NT AUTHORITY\SYSTEM	services.exe	WinDefend Service Stopped
2026-03-30 01:06:45	WIN-SOC-PROD-04	WIN-SOC\SvcOps	powershell.exe	AMSI Bypass via Reflection Patching
2026-03-30 01:07:30	WIN-SOC-PROD-04	WIN-SOC\SvcOps	reg.exe	ETW Provider Disabled via Registry to Suppress Logging
2026-03-30 02:00:05	WIN-SOC-PROD-05	WIN-SOC\NetAdmin	net.exe	net view /domain used for Domain Enumeration
2026-03-30 02:00:30	WIN-SOC-PROD-05	WIN-SOC\NetAdmin	nltest.exe	nltest /dclist used for Domain Controller Enumeration
2026-03-30 02:01:00	WIN-SOC-PROD-05	WIN-SOC\NetAdmin	arp.exe	ARP Cache Scan for Local Network Host Discovery

XXI

ANNEXURE C - GLOSSARY OF TERMS

MITRE ATT&CK;	Knowledge base of adversary tactics and techniques used for threat attribution.
EID	Windows Event ID representing a specific audit event category.
Defense Evasion	Adversary behavior intended to avoid detection and disrupt investigation.
Lateral Movement	Use of compromised accounts/processes to move across hosts.
Persistence	Mechanisms that allow attacker presence after reboot/logoff.
Risk Index	Weighted score from event severity, behavior, and trust indicators.